Answer Key — Our Home: Earth, a Unique Life-Sustaining Planet

Final answers with brief reasons.

Objective

Q1 **(a) right distance from the Sun** — the right distance keeps water liquid.

Q2 **(b) Venus** — Venus, due to its thick CO_2 atmosphere.

Q3 **(a) UV rays** — ozone blocks harmful UV rays.

Q4 **(a) cosmic rays and solar wind** — it deflects cosmic rays and the solar wind.

Q5 **(b) photosynthesis** — photosynthesis releases oxygen.

Q6 **(b) False** — a mild greenhouse effect keeps Earth warm enough for life.

Q7 **habitable** — the habitable (Goldilocks) zone.

Q8 **70** — about 70% is water.

Short answer

Q9 It is just right so water stays liquid — not too hot (evaporates) nor too cold (freezes) — and liquid water is essential for life.

Q10 The ozone layer blocks harmful UV rays; the magnetic field deflects cosmic rays and the solar wind, protecting the atmosphere and life.

Q11 Photosynthesis takes CO_2 and releases O_2; respiration uses O_2 and releases CO_2 — together they recycle the two gases.

Long / application

Q12 (1) Right distance from the Sun — keeps water liquid (habitable zone). (2) Right size — gravity holds an atmosphere with oxygen and an ozone layer that blocks UV. (3) Magnetic field from molten iron in the core — shields Earth from harmful cosmic rays and solar wind.

Q13 Venus has a very thick carbon dioxide atmosphere that traps heat strongly (a runaway greenhouse effect), so it is hotter than Mercury even though Mercury is closer to the Sun. On Earth a mild greenhouse effect is helpful — it keeps us warm enough for liquid water — but it warns that too much greenhouse gas can overheat a planet, which is why rising CO_2 is a concern.

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